1) If 2π is the period of $\sin x$ then what will be the value of $\sin(x + 16\pi)$?

(a) $\sin x$ (b) $\cos x$

(c) $-\cos x$ (d) $-\sin x$

2) Using the method of variation of parameters, the particular integral for $y'' + y = cosec\ x$ is

(a) $-x\ sin\ x + \cos x\,(\log \sin x)$ (b) $-xcos\ x + \sin x\,(\log \sin x)$

(c) $cos\ x + \sin x\,(\log \sin x)$ (d) $cos\ x + \sin\ (\log \sin x)$

3) The equation $\dfrac{dx}{dy} + Px = Q$ is a linear differential equation if

(a) P and Q are function of y and constant only (b) P and Q are function of x and constant only

(c) P is function of x and Q a function of y^2 (d) P is function of y and Q a function of x^2

4) The value of the Fourier coefficient b_n for $f(x) = x^3$; $-\pi < x < \pi$ will be

(a) $2(-1)^n\left[\dfrac{-\pi^2}{n} + \dfrac{6}{n^3}\right]$ (b) $2(-1)^n\left[\dfrac{\pi^2}{n} + \dfrac{6}{n^3}\right]$

(c) $2\left[\dfrac{-\pi^2}{n} + \dfrac{6}{n^3}\right]$ (d) $-2\left[\dfrac{-\pi^2}{n} + \dfrac{6}{n^3}\right]$

5) Fourier transform of $f(x) = 1 - x^2\ if\ |x| \le 1$ and $fx(x) = 0, if\ |x| > 1$

(a) $\dfrac{4}{\sqrt{2\pi}}[-s\cos s + \sin s]$ (b) $\dfrac{4}{s^3\sqrt{2\pi}}[-s\cos s + \sin s]$

(c) $\dfrac{4}{s^3\sqrt{2\pi}}[s\cos s + \sin s]$ (d) $\dfrac{4}{s^3\sqrt{2\pi}}[\cos s + s\sin s]$

6) The order and degree of the given differential equation $\dfrac{d^2y}{dx^2} + \sqrt{1 + \left(\dfrac{dy}{dx}\right)^3} = 0$ is

(a) 1, 3 (b)2, ½

(c) 2,1 (d) 2, 2

7) The Fourier cosine transform of $f(x) = e^{-2x} + 4e^{-3x}$ will be

(a) $\sqrt{\dfrac{2}{\pi}}\left[\dfrac{1}{4+s^2} - \dfrac{6}{9+s^2}\right]$ (b) $\sqrt{\dfrac{2}{\pi}}\left[\dfrac{2}{4+s^2} + \dfrac{6}{9+s^2}\right]$

(c) $2\sqrt{\dfrac{2}{\pi}}\left[\dfrac{1}{4+s^2} + \dfrac{6}{9+s^2}\right]$ (d) $\sqrt{\dfrac{2}{\pi}}\left[\dfrac{4}{4+s^2} + \dfrac{6}{9+s^2}\right]$

8) $f(x,y) = \cos x + e^x + y$ is

(a) a homogeneous function of degree 1 (b) non - homogeneous function

(c) a homogeneous function of degree 2 (d) a homogeneous function of degree 0

9) The integrating factor for the Leibnitz linear differential equation $\frac{dz}{dy} - 4z = 2y^2$ is given by:

(a) $e^{\int -4dy}$ (b) $e^{\int -4dz}$

(c) $e^{\int 4dy}$ (d) $e^{\int 4dz}$

10) For the given differential equation $(2x+y)dx = (-x-y)dy$, check whether

(a) $\frac{\partial M}{\partial x} = \frac{\partial N}{\partial y}$ (b) $\frac{\partial M}{\partial y} + \frac{\partial N}{\partial x} = 0$

(c) $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ (d) $\frac{\partial^2 M}{\partial y^2} = \frac{\partial N}{\partial x}$

11) The value of wronskian $W\left(1, x, x^2\right)$ is

(a) $2x^4$ (b) 2

(c) 0 (d) 1

12) The particular integral for $y'' - y = 2x^4 - 3x + 1$ will be

(a) $-2x^4 - 24x^2 + 3x + 49$ (b) $-2x^4 - 24x^2 + 3x - 49$

(c) $2x^4 - 24x^2 + 3x - 49$ (d) $2x^4 + 24x^2 + 3x - 49$

13) If z_1 and z_2 are two real and equal roots of second order homogeneous differential equation then the solution will be (c_1 and c_2 are constants)

(a) $y = (c_1 + xc_2)\mathrm{e}^{z_1 x}$ (b) $y = (c_1\mathrm{e}^{z_1 x} + c_2\mathrm{e}^{z_2 x})$

(c) $y = (c_1 + \mathrm{c}_2)\mathrm{xe}^{z_1 x}$ (d) $y = \mathrm{e}^{z_1 x}(\cos z_2 x + \sin z_2 x)$

14) If $(1+xy)\,\mathrm{ydx} + (1-\mathrm{xy})\,\mathrm{xdy} = 0$ then the integrating factor will be

(a) $\frac{1}{2xy}$ (b) $\frac{1}{2x^2y^2}$

(c) $\frac{1}{2xy} + \frac{1}{2x^2y^2}$ (d) $-\frac{1}{2x^2y^2}$

15) If $f(x) = |x|$ is defined over $[0, 2\pi]$ then a_0 will be

(a) 9 π/2 (b) 2 π

(c) π/2 (d) π

16) If the roots of the differential equation are 2,2,3, then the solution will be

(a) $y = (Ax + x^2 B)\mathrm{e}^{2x} + C\,\mathrm{e}^{3x}$ (b) $y = (Ae^{3x} + xB)\mathrm{e}^{3x} + C\,\mathrm{e}^{2x}$

(c) $y = (A + xB)\mathrm{e}^{2x} + C\,\mathrm{e}^{3x}$ (d) $y = (\mathrm{A} + x^2\mathrm{e}^{3x} B) + C\,\mathrm{e}^{2x}$

17) The value of Fourier coefficient a_n for $f(x) = e^x$, $0 < x < 1$ half range cosine is given by

(a) $\frac{2}{1+n^2\pi^2}[(-1)^n e - 1]$ (b) $\frac{2}{1-n^2\pi^2}[(-1)^n e - 1]$

(c) $\frac{2}{1+n^2\pi^2}[e - 1]$ (d) $\frac{2}{1+n^2\pi^2}[(-1)^n e + 1]$

18) The integrating factor for $\left(xy^2 - e^{\frac{1}{x^2}}\right)dx - x^2ydy = 0$ will be

(a) x^4 (b) $\dfrac{4}{x}$

(c) x^{-4} (d) $\dfrac{-4}{x}$

19) The orthogonal trajectories for the one-parameter family of curve $y^2 = cx$ will be given by (c is a constant)

(a) $\dfrac{x^2}{2} + \dfrac{y^2}{2} = c$ (b) $x^2 + \dfrac{y^2}{2} = c$

(c) $x^3 + y^2 = c$ (d) $\dfrac{x^2}{2} - \dfrac{y^2}{2} = c$

20) The solution of $\dfrac{dy}{dx} = \dfrac{x^2 + y^2}{xy}$ is given by

(a) $\log x^2 + \dfrac{y^2}{2x^2} = c$ (b) $\log x + \dfrac{y^2}{2x^2} = c$

(c) $\log x - \dfrac{y}{2x} = c$ (d) $\dfrac{y^2}{2x^2} - \log x = c$

21) Complementary function for $y'' + 4y = 0$ will be (A and B are constants)

(a) $y = A\cos\sqrt{2}x + B\sin\sqrt{2}x$ (b) $y = Ax\cos 2x + Bx\sin 2x$

(c) $y = A\cos 2x + B\sin 2x$ (d) $y = Ax\cos\sqrt{2}x + Bx\sin\sqrt{2}x$

22) The particular integral for the non-homogeneous equation $(\mathrm{D}^3 - 3D^2 + 4D - 2)y = e^x + \cos x$ is given by:

(a) $xe^x + \dfrac{1}{10}(3\sin x + \cos x)$ (b) $e^x - \dfrac{2}{10}(3\cos x + \sin x)$

(c) $e^x + x + \dfrac{1}{10}(3\cos x + \sin x)$ (d) $xe^x + \dfrac{1}{10}(3\sin x - 3\cos x)$

23) Fourier sine transform of $f(x) = \dfrac{1}{x}$ is given by

(a) 1 (b) 0

(c) $\sqrt{\pi/2}$ (d) π

24) The solution of the Bernoulli equation $(xy^5 + y)dx = dy$ is given by (c is a constant)

(a) $y^{-4}e^{-4x} = -xe^{-4x} + \dfrac{e^{-4x}}{4} + c$ (b) $y^4e^{-4x} = -xe^{4x} + \dfrac{e^{4x}}{4} + c$

(c) $y^{-4}e^{4x} = -xe^{4x} + \dfrac{e^{4x}}{4} + c$ (d) $y^4e^{4x} = +xe^{4x} - \dfrac{e^{4x}}{4} + c$

25) What will be the orthogonal trajectory for $e^x + e^{-y} = c$? (A and c are constants)

(a) $e^x + e^{-y} = A$ (b) $e^y - e^{-x} = A$

(c) $e^x - e^{-y} = c$ (d) $-e^x + 2e^{-y} = c$

26) Fourier coefficient a_n for $f(x) = x+x^2$ over the interval [-π, π] will be

(a) $\frac{4(-1)^n}{n^2}$ (b) $\frac{4(-1)^{2n}}{n^2}$

(c) $\frac{4(-1)^3}{n^3}$ (d) $\frac{4(-1)^n}{\pi^2}$

27) If the Fourier series representation for $f(x) = x^2, \ -\pi < x < \pi$ is given by

$f(x) = \frac{\pi^2}{3} - 4\left[\frac{\cos x}{1^2} - \frac{\cos 2x}{2^2} + \frac{\cos 3x}{3^2} - \frac{\cos 4x}{4^2} + \ldots\right]$ then $\left[\frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \ldots\right]$ will be equal to

(a) $\frac{\pi}{12}$ (b) $\frac{\pi^2}{3}$

(c) $\frac{\pi^2}{4}$ (d) $\frac{\pi^2}{12}$

28) The complete solution for $p - \frac{1}{p} = \frac{x}{y} - \frac{y}{x}$

(a) $(y \log x - c)(x^2 - y^2 - c)$ (b) $(xy - c)(x^2 - y^2 - c)$

(c)) $(xy - c)(x^2 + y^2 - c)$ (d)) $(x^2y^2 - c)(x - y - c)$

29) The general solution of the given differential equation $p = \cos(y - xp)$ is (c is a constant)

(a) $p = \sin(y - xp)$ (b) $y = cx + \cos^{-1} c$

(c) $c = \cos(c - xy)$ (d) $y = \sec(x - xc)$

30) The value of the Fourier coefficient b_n for $f(x) = x \sin x$; $-\pi < x < \pi$ will be

(i) 0 (ii) $\frac{1}{\pi}\int_0^{\pi} x \sin x \sin nx dx$

(v) $\frac{2}{\pi}\int_{-\pi}^{\pi} x \sin x \sin nx dx$ (iv) $\frac{2}{\pi}\int_0^{\pi} x \sin x \cos nx dx$

31) The order of the given differential equation $xy\frac{d^2y}{dx^2} + x\left(\frac{dy}{dx}\right)^2 - y\frac{dy}{dx} = 0$ is

(i) 1 (ii) 2

(iii) 0 (iv) ½

32) The function $f(x) = 3x^3+xy^2+\sin x+ e^x$ is

(i) a homogeneous function of degree 1 (ii) not a homogeneous function

(iii) a homogeneous function of degree 2 (iv) a homogeneous function of degree 3

33) The value of wronskian $W\left(x, x^2, x^3\right)$ is

(i) $2x^4$ (ii) $2x^3$

(iii) 0 (iv) 1

34) For the given differential equation $(2x + y)dx = (-x - y)dy$, check whether

a) $\frac{\partial M}{\partial x} = \frac{\partial N}{\partial y}$ b) $\frac{\partial M}{\partial y} + \frac{\partial N}{\partial x} = 0$

c) $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ d) $\frac{\partial^2 M}{\partial y^2} = \frac{\partial N}{\partial x}$

35) Bernoulli's equation $y' + P(x)y = Qy^n$ can be reduces to linear differential equation is

a) $\frac{du}{dx} + nP(x)u = Q(x)$ $(b)\frac{du}{dx} + (1-n)P(x)u(1-n)Q(x)$

c) $\frac{du}{dx} + P(x)u = nQ(x)$ d) $\frac{du}{dx} + nP(x)u = (1+n)Q(x)$

36) The general solution of the equation $\frac{dy}{dx} = \frac{-x}{y}$ is

a) $x^2y^2 = a^2$ b) $x + y = a^2$

c) $x^2 - y^2 = a^2$ d) $x^2 + y^2 = a^2$

37) Differential equation xdy+ydx=

a) d(xy) b) d(x-y)

c) d(x+y) d) d(x/y)

38) Differential equation $\frac{ydx-xdy}{y^2}$=

a) d(xy) b) d(x-y)

c) d(x+y) d) d(x/y)

39) Differential equation $\frac{xdy-ydx}{x^2+y^2}$=

a) d(xy) b) d(x-y)

c) $d(\tan^{-1}\frac{y}{x})$ d) $d(\tan^{-1}\frac{x}{y})$

40) If differential equation is of the form $f_1(xy)ydx + f_2(xy)xdy = 0$ then Integrating Factor is

(a) $\frac{1}{Mx-Ny}$ provided $Mx - Ny \neq 0$

(b) $\frac{1}{Mx+Ny}$ provided $Mx + Ny \neq 0$

(c) $\frac{1}{My-Nx}$ provided $My - Nx \neq 0$

(d) $\frac{1}{My+Nx}$ provided $My + Nx \neq 0$.

41) If $f(x, y, \frac{dy}{dx}) = 0$is a differential equation of family curves $F(x, y, c) = 0$. Then orthogonal trajectories is

(a) $f(x, y, \frac{dy}{dx}) = 0$

(b) $f(x, y, -\frac{dy}{dx}) = 0$

(c) $f(-x, -y, \frac{dy}{dx}) = 0$

(d) $f(x, y, \frac{-dx}{dy}) = 0$.

42) Two family of curves cuts each member of other family at right angles are called……….

(a) Normal trajectories

(b) Orthogonal angle

c) Orthogonal trajectories

d) Circle

43) The equation $ydx + xdy = 0$ is-

a) Exact differential equation

b)No-exact differential equation

c) Partial differential equation

d) None of these.

44) If $Mdx + Ndy = 0$ is a differential equation and $\frac{1}{M}\left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right)$ is a function of y alone, say $f(y)$, then integrating factor is

(a) e^x

b) e^{x+y}

(c) $e^{\int f(y)dy}$

d) $e^{-\int f(y)dy}$

45) If an equation $Mdx + Ndy = 0$ is not exact. Let $F(x, y)$ be such that $FMdx + FNdy = 0$ is exact, then the function F is called-

a) Differentiable function

b) Differentiable factor

c) Integrating factor

d) Arbitrary function.

46) The solution of equation $ydx + xdy = 0$ is-

(A) $xy = c$

(b) $x = y + c$

(c) $x - y = c$

(d) None of these.

47) If equation $Mdx + Ndy = 0$ is homogeneous differential equation, then I.F. is

(a) $\frac{1}{Mx-Ny}$ provided $Mx - Ny \neq 0$

(b) $\frac{1}{Mx+Ny}$ provided $Mx + Ny \neq 0$
(c) $\frac{1}{My-Nx}$ provided $My - Nx \neq 0$
(d) $\frac{1}{My+Nx}$ provided $My + Nx \neq 0$.
48) If the roots of the differential equation are 2,2,3, then the solution will be
(a) $y = (Ax + x^2B)\, e^{2x} + Ce^{3x}$
(b) $y = (Ae^{3x} + xB)\, e^{3x} + Ce^{2x}$
(c) $y = (A + xB)\, e^{2x} + Ce^{3x}$
(D) $y = (A + x^2\, e^{3x}\, B) + Ce^{2x}$

50) Complementary function for $y'' + 4y = 0$ will be
(A and B are constants)
(a) $y = A\, cos\, \sqrt{2}\, x + B\, sin\, \sqrt{2}\, x$
(b) $y = Ax\, cos\, 2\, x + Bx\, sin\, 2\, x$
(c) $y = A\, cos\, 2\, x + B\, sin\, 2\, x$
(d) $y = Ax\, cos\, \sqrt{2}\, x + Bx\, sin\, \sqrt{2}\, x$

51) The Particular integral for the non-homogeneous equation $(D^3 - 3D^2 + 4D - 2)y = e^x + cos\, x$ is given by:
(a) $xe^x + \frac{1}{10}(3\, sin\, x + cos\, x)$
(b) $e^x - \frac{2}{10}(3\, cos\, x + sin\, x)$
(c) $e^x + x + \frac{1}{10}(3\, cos\, x + sin\, x)$
(d) $xe^x + \frac{1}{10}(3\, sin\, x - 3\, cos\, x)$

52) The differential equation $xyp^2 + p(3x^2 - 2y^2) - 6xy = 0$ is solvable by
a) p
b) x
c) y
d) x and y both

53) Which of the following a solution of differential equation $y = px + \frac{a}{p}$

a $y = cx^2 + \frac{a}{c}$
b $y = cx + \frac{a}{c}$
c $y = x + \frac{a}{c}$
d None of these

54) The differential equation $3x^4p^2 - xp - y = 0$ is solvable by
a) p
b) x
c) y
d) x and $y$both

55) The differential equation $p^2 + xp + y = 0$ is solvable by

a) p b) y

c) x d) x and y both

56) Solution of the O.D.E. $y=xp + p^2+ 2$ is

a) $y=xc + p^2+ 2$ b) $y=xc + c^2$

c) $y=xc + c^2+ 2$ d) $y=xp + c^2+ 2$

57) The complementary function of the differential equation $(D^2 - 5D + 6)y = e^{4x}$ is:

a) $y_c = c_1 e^{-2x} + c_2 e^{3x}$ b) $y_c = c_1 e^{-2x} + c_2 e^{-3x}$

c) $y_c = c_1 e^{2x} + c_2 e^{-3x}$ d) $y_c = c_1 e^{2x} + c_2 e^{3x}$

58) Complimentary function is the general solution of:

(a) Homogeneous Linear Differential Equations

(b) Non-Homogeneous Linear Differential Equations

(c) Non-Homogeneous and Homogeneous Linear Differential equations

(d) Non-Linear Differential Equations

59) If m_1, m_2, m_3……. are real and different roots then complimentary function is:

(A) $(c_1 + c_2)e^{m_2 x} + c_3 e^{m_3 x}+$…….

(b) $(c_1 + c_2 x)e^{m_2 x} + c_3 e^{m_3 x}+$…….

(c) $c_1 e^{m_1 x} + c_2 e^{m_2 x} + c_3 e^{m_3 x}+$…….

(d) $c_1 e^{m_1 x} + c_2 e^{m_2 x} + c_3 e^{m_3 x}$

60) What is the complimentary function if auxiliary equation has roots as 2, 2 and 3

(a) $(c_1 x + c_2)e^{2x} + c_3 e^{3x}$

(b) $(c_1 + c_2)e^{2x} + c_3 e^{3x}$

(c) $(c_1 x + c_2)e^{-2x} + c_3 e^{3x}$

(d) $(c_1 x + c_2)e^{2x} + c_3 e^{-3x}$

61) What is the complimentary function if auxiliary equation has roots as 2, -3, -3 and -3.

(A) $(c_1 x^2 + c_2 x + c_3)e^{-3x} + c_4 e^{2x}$

(b) $(c_1 x^2 + c_2 x + c_3)e^{3x} + c_4 e^{2x}$

(c) $(c_1 x^2 + c_2 x + c_3)e^{-3x} + c_4 e^{-2x}$

(d) $(c_1 x^2 + c_2 x + c_3)e^{-3x} + e^{2x}$

62)What is the complimentary function if auxiliary equation has roots as $2\pm3i$ and -3.

(a) $e^{2x}(c_1 \cos 3x + \sin 3x) + c_3 e^{-3x}$

(b) $e^{2x}(c_1 \cos 3x + c_2 \sin 3x) + c_3 e^{-3x}$

(c) $e^{2x}(\cos 3x + c_2 \sin 3x) + c_3 e^{-3x}$

(d) $e^{2x}(c_1 \cos 3x + c_2 \sin 3x) + e^{-3x}$

63)What is the complimentary function if auxiliary equation has roots $a \pm i\beta$,$a \pm i\beta$ and -3.

(a) $e^{ax}[(c_1 + c_2)\cos\beta x + (c_3 x + c_4)\sin\beta x] + c_5 e^{m_5 x}$

(b) $e^{ax}[(c_1 x + c_2)\cos\beta x + (c_3 + c_4)\sin\beta x] + c_5 e^{m_5 x}$

(c) $[(c_1x+c_2)\cos\beta\, x+(c_3x+c_4)\sin\beta\, x]+c_5e^{m_5x}$

(d) $e^{ax}[(c_1x+c_2)\cos\beta\, x+(c_3x+c_4)\sin\beta\, x]+c_5e^{-3x}$

64) $y_p=\frac{1}{D-a}e^{ax}$=?

(a) $-xe^{ax}$

(b) xe^{ax}

(c) xe^{x}

(d) e^{ax}

65)$y_p=\frac{1}{(D-a)^n}e^{ax}$ =?

(a) $\frac{x^2}{n!}e^{ax}$

(b) $\frac{x^n}{n!}e^{x}$

(c) $\frac{x^n}{n!}e^{ax}$

(d)x^ne^{ax}

66) $y_p=\frac{1}{D^2+a^2}\sin(ax+b)$ =?

(a) $\frac{x\cos(ax+b)}{(2a)}$

(b) $-\frac{\cos(ax+b)}{(2a)}$

(c) $-\frac{x\cos(ax)}{(2a)}$

(d)$-\frac{x\cos(ax+b)}{(2a)}$

68) $y_p=\frac{1}{(D^2+a^2)^n}\sin(ax)$ =?

(a) $\frac{x^n}{(2a)^n n!}\sin\left(ax-\frac{n\pi}{2}\right)$

(b) $\frac{x^n}{(2a)^n n!}\sin\left(ax+\frac{n\pi}{2}\right)$

(C) $\frac{x^n}{(2a)^n}\sin\left(ax-\frac{n\pi}{2}\right)$

(d) $\frac{1}{(2a)^n n!}\sin\left(ax-\frac{n\pi}{2}\right)$

69) $y_p=\frac{1}{(D^2+a^2)^n}\cos(ax+b)$ =?

(a) $\frac{1}{(2a)^n n!}\cos\left(ax-\frac{n\pi}{2}\right)$

(b)$\frac{x^n}{(2a)^n n!}\cos\left(ax-\frac{n\pi}{2}\right)$

(c) $\frac{x^n}{(2a)^n}\cos\left(ax-\frac{n\pi}{2}\right)$

(d) $\frac{x^n}{(2a)^n n!}\cos\left(ax+\frac{n\pi}{2}\right)$

70) $y_p=\frac{1}{f(D)}e^{ax}V(x)$ =?

(a) $e^{ax}\frac{1}{f(D+a)}V(x)$

(b) $e^{ax}\frac{1}{f(D-a)}V(x)$

(c) $e^{ax}V(x)\frac{1}{f(D+a)}$

(d) $V(x)\frac{1}{f(D+a)}e^{ax}$

71) $y_p = \frac{1}{f(D)}e^{ax+b}$ =?

(a) $y_p = \frac{1}{f(a)}e^{ax+b}$

b) $y_p = \frac{-1}{f(a)}e^{ax+b}$

c) $y_p = \frac{x}{f(a)}e^{ax+b}$

d) $y_p = \frac{1}{f(a)}e^{ax}$

72) An equation of the form $y = px + f(p)$ is known as………… equation.

a)Bernoulli's

b)Claiuaut's

c)Leibnitz's

d)Legendre's.

73) The complete solution of non-homogeneous differential equations is given as:

a)The sum of complimentary function and complimentary integral.

b)The sum of particular function and complimentary integral.

c)The sum of particular function and particular integral.

d) The sum of complimentary function and particular integral.

74)The C.F. of the differential equation $(D^2 + 1)y = \sin x$ is

(a) $(a_1 \cos x + a_2 \sin x)$

(b) $(a_1 \cos x + a_2 \sin x)x$

(c) $(a_1 \cos x + a_2 x \sin x)$

(d) $(a_1 \cos x + a_2 \sin x) \sin x$

75) The solution of differential equation contains as many arbitrary constants as the order of a differential equation, and then the solution is said-

a) Particular solution

b) Complete primitive

c) Singular solution

d) None of these

76) The Solution of the differential equation is

$(x^2 - ay)dx = (ax - y^2)dy$ is

a) $x^3 - 3axy + y^3 = c$ b) $5x^3 - 3axy = c$

c) $x^3 - 5axy + 2y^3 = c$ d) $x^3 + y^3 = c$

77) Which of the following is an integrating factor for the differential equation $(y^2 - xy)dx + x^2dy = 0$

a) $\frac{1}{xy^2}$ b) $\frac{1}{x^2y^2}$

c) $\frac{1}{y^2}$ d) $\frac{1}{y^3}$

78) If $(1 + xy)\, ydx + (1 - xy)\, xdy = 0$ then the integrating factor will be

a) $\frac{1}{2xy}$ b) $\frac{1}{2x^2y^2}$

c) $\frac{1}{2xy} + \frac{1}{2x^2y^2}$ d) $-\frac{1}{2x^2}$

79) The I.F. for $\left(xy^2 - e^{\frac{1}{x^2}}\right)dx - x^2ydy = 0$ will be

a) x^4 b) $\frac{4}{x}$

c) x^{-4} c) $\frac{-4}{x}$

80)The orthogonal trajectories for the one-parameter family of curve$y^2 = cx$ will be given by (*c* is a constant)

a) $\frac{x}{2} + \frac{y^2}{2} = c$ b) $x^2 + \frac{y^2}{2} = c$

c) $x^3 + y^2 = c$ d) $\frac{x^2}{2} - \frac{y^2}{2} = c$

81) The solution of $\frac{dy}{dx} = \frac{x^2+y^2}{xy}$ is given by

a) $\log x^2 + \frac{y^2}{2x^2} = c$ b) $\log x + \frac{y^2}{2x^2} = c$

c) $\log x - \frac{y}{2x} = c$ d) $\frac{y^2}{2x^2} - \log x = c$

82)Differential equation of circles $x^2 + y^2 = a^2$ with centre at origin is

a) $y\frac{dy}{dx} - x = 0$ b) $\frac{dy}{dx} + xy = 0$

c) $y\frac{dy}{dx} + x = 0$ d) $y\frac{dy}{dx} + xy = 0$

83) I.F. of differential equation $(xy^3 + y)dx + 2(x^2y^2 + x + y^4)dy = 0$ is

(A) $-y$

(b) y

(c) $\frac{1}{y}$

(d) $-\frac{1}{y}$.

84) Solution of differential equation is $e^y\, dx + (2y + x\, e^y)\, dy = 0$ is

(a) $xe^y + y^2 = c$

(b) $e^y + y^2 = c$

(c)$xe^y - y^2 = c$

(d) $e^y - y^2 = c.$

85) Solution of $(y-px)(p-1)=p$ is
(a) $(y-cx)(c-1)=c$
(b) $(y-cx)=0$
(c) $xy=c^2y+c$
(d) $x=c^2x+c.$

86) The particular integral of $(D^2+a^2)y=\sin ax$ is
a) $\frac{x}{2a}\cos ax + x$
b) $-\frac{ax}{2}\cos x$
c) $-\frac{x}{2a}\cos ax$
d) $\frac{ax}{2}\cos x$

87) The particular integral of the differential equation $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = \cos 3x$ is given by
(a) $-\frac{3}{25}\sin 3x$
(b) $-\frac{3}{25}\cos 3x$
(c) $\frac{1}{50}(-3\sin 3x - 4\cos 3x)$
(d) $\frac{1}{25}(-3\sin 3x + 4\cos 3x)$

88) $y_p = \frac{1}{f(D)}xV(x)$ =?
(a) $x\frac{1}{f(D)}V(x) + \frac{f'(D)}{[f(D)]^2}V(x)$
b) $x\frac{1}{f(D)}V(x) - \frac{1}{[f(D)]^2}V(x)$
c) $x\frac{1}{D}V(x) - \frac{f'(D)}{[f(D)]^2}V(x)$
d) $x\frac{1}{f(D)}V(x) - \frac{f'(D)}{[f(D)]^2}V(x)$

89) Solution of $\frac{dy}{dx} = log(x\frac{dy}{dx} - y)$ is given by
a) $y = cx^2 - e^c$
b) $y = cx - e^c$
c) $y = cx^2 + e^c$
d) $y = cx + e$

90) The complementary function for the differential equation $\frac{d^2y}{dx^2} - 8\frac{dy}{dx} + 15y = 0$ is
a) $c_1e^{3x} + c_2e^{x}$
b) $c_1e^{3x} + c_2e^{-5x}$
c) $c_1e^{-3x} + c_2e^{5x}$
d) $c_1e^{3x} + c_2e^{5x}$

91) The Particular integral for $\frac{d^2y}{dx^2} + 16y = sin\,4\,x$ is given by

a) $\frac{x}{8} cos\, 4\, x$

b) $\frac{-x}{10} sin\, 4\, x$

c) $\frac{-x}{8} cos\, 4\, x$

d) $\frac{-x}{10} cos\, 4\, x$

92) The complete solution for $p - \frac{1}{p} = \frac{x}{y} - \frac{y}{x}$

a) $(y \log x - c)(x^2 - y^2 - c) = 0$

b) $(xy - c)(x^2 - y^2 - c) = 0$

c) $(xy - c)(x^2 + y^2 - c) = 0$

d) $(x^2y^2 - c)(x - y - c) = 0$

93) The solution of the differential equation $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + 5y = 0$ is

a) $e^{2x}(c_1\, cos\, x + c_2\, sin\, x)$

b) $e^{x}(c_1\, cos\, 2\, x + c_2\, sin\, 2\, x)$

c) $e^{-2x}(c_1\, cos\, x + c_2\, sin\, x)$

d) $e^{-x}(c_1\, cos\, 2\, x + c_2\, sin\, 2\, x)$

94) In the Fourier series expansion of a periodic function $f(x) = \frac{1}{2}(\pi - x), 0 < x < 2\pi$, the coefficient b_n is

(a) π

(b) $\frac{1}{n}$

(c) $\frac{\pi}{2}$

(d) 0

95) If in a Fourier series expansion of a periodic function $f(x) = \frac{3x^2 - 6x\pi + 2\pi^2}{12}$, $0 < x < 2\pi$, the Fourier coefficients are $a_0 = 0$, $a_n = \frac{1}{n^2}$, $b_n = 0$ then

(a) $\frac{\pi^2}{6} = 1 + \frac{1}{2^2} + \frac{1}{3^2} + ..$

(b) $\frac{\pi^2}{6} = 1 + \frac{1}{2^2} - \frac{1}{3^2} - \cdots$

(c) $\frac{\pi^2}{6} + 1 - \frac{1}{2^2} + \frac{1}{3^2} + ..$

(d) $\frac{\pi^2}{6} = 1 - \frac{1}{2^2} - \frac{1}{3^2} - \cdots$

96) The coefficient b_3 in a Fourier series expansion of a periodic function $f(x) = \begin{cases} -1, & 0 < x < \pi \\ 2, & 0 < x < 2\pi \end{cases}$ is

(a) $-\frac{2}{\pi}$

(b) 1

(c) $\frac{\pi}{2}$

(d) 0

97) The coefficient a_2 in a Fourier series expansion of a periodic function $f(x) = x^2, 0 < x < 4$ is

(a) $\frac{8}{\pi^2}$

(b) $\frac{4}{\pi^2}$

(c) $\frac{16}{\pi^2}$

(d) $\frac{4}{\pi}$

98) If the $f(x)$ is even then

(a) $\int_{-l}^{l} f(x)dx = \int_0^l f(x)dx$

(b) $\int_{-l}^{l} f(x)dx = 2\int_0^l f(x)dx$

(c) $\int_{-l}^{l} f(x)dx = 0$

(d) $\int_{-l}^{l} f(x)dx = 1$

99) In the Fourier cosine series expansion of a periodic function $f(x) = 1, 0 < x < \pi$, the coefficient a_n is

(a) $\frac{2}{\pi}\int_0^{\pi} \sin nx\, dx$

(b) $\frac{2}{\pi}\int_0^{\pi} \cos nx\, dx$

(c) $\frac{1}{\pi}\int_0^{\pi} \sin nx\, dx$

(d) $\frac{1}{\pi}\int_0^{\pi} \sin nx\, dx$

100)The value of the Fourier coefficient b_n for $f(x) = x^3 \sin x$; $-\pi < x < \pi$ will be

(a) 0

(b) $\frac{1}{\pi}\int_0^{\pi} x^3 \sin x \sin nx dx$

(c) $\frac{2}{\pi}\int_{-\pi}^{\pi} x^3 \sin x \sin nx dx$

(d) $\frac{2}{\pi}\int_0^{\pi} x^3 \sin x \cos nx dx$

101) Which of the following functions is an even function in the interval $[-\pi, \pi]$?

(a) x

(b) $x^3 \sin x$

(c) $x^3 + \cos x$

(d) $x^3 + \sin x$

102) In the Fourier series expansion of the periodic function $f(x) = x$, $0 < x < 2\pi$ the coefficient of $\sin 2x$ is

(a) 2

(b) π

(c) 0

(d) -1

103) If $f(x) = x^2 + x^4$ in $-\pi \le x \le \pi$, then b_n is _______.

(a) $\frac{2}{\pi}$

(b) $\frac{2}{\pi}$

(c) 0

(d) $\frac{2}{\pi}$

104) If the Fourier transform of the function f(x) is F[S], then the Fourier transform of the function f(3x) is

(a) $e^{2is} F\left[\frac{s}{3}\right]$

(b) $e^{3s} F\left[\frac{s}{3}\right]$

(c) $3F\left[\frac{s}{3}\right]$

(d) $\frac{1}{3} F\left[\frac{s}{3}\right]$

105) If the Fourier series expansion of the periodic function $f(x) = \begin{cases} -\frac{\pi}{4}, -\pi < x < 0 \\ \frac{\pi}{4}, 0 < x < \pi \end{cases}$ is given by $\frac{\sin x}{1} + \frac{\sin 3x}{3} + \frac{\sin 5x}{5} + \frac{\sin 7x}{7} + \cdots$ then the series $1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \ldots$ converges to

(a) $\frac{\pi}{8}$

(b) $\frac{\pi}{4}$

(c) 0

(d) $-\frac{\pi}{4}$

106) The coefficient function of the dependent variable in the linear differential equation corresponding to the Bernoulli equation $\frac{dy}{dx} + \frac{y}{x} = x^2y^2$ is

(a) $-x^{-1}$ (b) x

(c) $-x$ (d) x^{-1}

107) The solution of differential equation $ylogy\ dx + (x - logy)dy = 0$ is

(a) $xlogy = \frac{(logy)^2}{3} + c$ (b) $xlogy = \frac{(logy)^2}{2} + c$

(c) $logy = \frac{(logy)^2}{2} + c$ (d) $logy = \frac{(logy)^2}{3} + c$

108) The order of differential equation $\left(\frac{d^2y}{dx^2} + 2x\frac{dy}{dx}\right)^{3/2} = 2x$ is

(a) 2 (b) 1

(c) 3 (d) 0

109) The integrating factor of the differential equation $(x+1)\frac{dy}{dx} - y = e^x(x+1)^2$ is

(a) $(x+1)$ (b) $\frac{1}{(x+1)^2}$

(c) $\frac{1}{(x+1)}$ (d) $(x+1)^2$

110) The integrating factor of the differential equation $(xlogx)\frac{dy}{dx} + y = (logx)^2$ is

(a) $logx$ (b) x

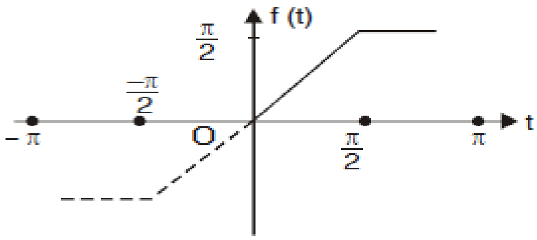
(c) $\frac{1}{(x+1)}$ (d) $\frac{1}{x}$

111) The Fourier Cosine transform of $f(x) = e^{-x}$ is $\sqrt{\frac{2}{\pi}}\left(\frac{1}{s^2+1}\right)$. Use Parseval's Identity to evaluate $\int_0^\infty \frac{dx}{(x^2+1)^2}$

(a) $\frac{\pi}{3}$ (b) $\frac{\pi}{2}$

(c) $\frac{\pi}{4}$ (d) π

112) The coefficient b_2 in Fourier sine series expansion of a periodic function f(t) given as below graph is



(a) $-\frac{2}{9\pi}+\frac{1}{3}$ (b) $-\frac{1}{2}$

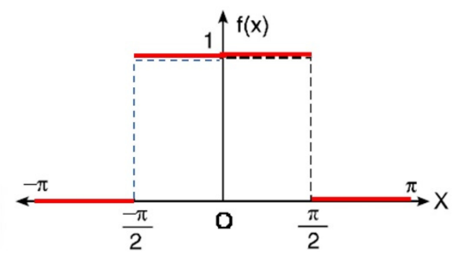
(c) $\frac{2}{\pi}+1$ (d) 0

. 113) The Fourier sine transform of $f(x) = e^{-2x}$ is __________.

(a) $\sqrt{\frac{2}{\pi}}\frac{s}{s^2+4}$ (b) $\sqrt{\frac{2}{\pi}}\frac{2}{s^2+4}$

(c) $\sqrt{\frac{2}{\pi}}\frac{s}{s^2+2}$ (d) $\sqrt{\frac{2}{\pi}}\frac{2}{s^2+2}$

114) The value of Euler constant a_0 in the Fourier series of the following function is _______.



(a) 2 (b) 1

(c) -1 (d) -2

115) The Fourier series for the function $f(x) = x^2 \; in -\pi < x < \pi$ is

(a) $f(x) = \frac{\pi^2}{3} + 2\sum_{n=1}^{\infty}\frac{(-1)^n}{n^2}cosnx$

(b) $f(x) = \frac{\pi^2}{2} + 4\sum_{n=1}^{\infty}\frac{(-1)^n}{n^2}cosnx$

(c) $f(x) = \frac{\pi^2}{3} - 4\sum_{n=1}^{\infty}\frac{(-1)^n}{n^2}cosnx$

(d) $f(x) = \frac{\pi^2}{3} + 4\sum_{n=1}^{\infty}\frac{(-1)^n}{n^2}cosnx$

116) A function f(x) is periodic with period 2π if:

A. f(x + 2π) = f(x)

B. f(x + π) = f(x)

C. f(−x) = f(x)

D. None of these

117) Fourier series of f(x) on (0, 2π) is:
A. a_0 + Σ (an cos nx + bn sin nx)
B. Σ (an cos nx + bn sin nx)
C. $\frac{a_0}{2}+\sum_{n=1}^{\infty}(a_n \cos nx + b_n \sin nx)$
D. Σ an sin nx

118) The coefficient a_0 equals:
A. $(1/\pi) \int_a^{a+2\pi} f(x)\, dx$
B. $(1/2\pi) \int_a^{a+2\pi} f(x)\, dx$
C. $\int_a^{a+2\pi} f(x)\, dx$
D. None of these

119) If f(x) is even, its Fourier series contains:
A. Only sine terms
B. Only cosine terms
C. Both sine and cosine terms
D. Exponential terms

120) If f(x) is odd, its Fourier series contains:
A. Only cosine terms
B. Only constant term
C. Only sine terms
D. None of these

121) Fourier Coefficient an on (−L, L) equals:
A. $(\frac{1}{L}) \int_{-L}^{L} \mathrm{f(x)} \cos(\frac{\mathrm{n\pi x}}{\mathrm{L}})\, \mathrm{dx}$
B. $(\frac{2}{L}) \int_{-L}^{L} \mathrm{f(x)} \cos(\frac{\mathrm{n\pi x}}{\mathrm{L}})\, \mathrm{dx}$
C. (1/π) ∫ f(x) cos nx dx
D. None

122) Half-range cosine series is used for:
A. Odd function
B. Even function
C. Non-periodic function
D. Discontinuous function

123) Half-range sine series is obtained by:
A. Even extension
B. Odd extension
C. Periodic extension

D. Constant extension

124) Coefficient bn in half-range sine series equals:
A. $(\frac{1}{L})\ \int_0^L f(x)\ sin(\frac{n\pi x}{L})\ dx$
B. $(\frac{4}{L})\ \int_0^L f(x)\ sin(\frac{n\pi x}{L})\ dx$
C. $(\frac{2}{L})\ \int_0^L f(x)\ sin(\frac{n\pi x}{L})\ dx$
D. None

125) Fourier transform of f(x) is defined as:
A. $\frac{1}{\sqrt{2\pi}}\int_0^\infty f(x)e^{-i\omega x}\ dx$
B. $\frac{1}{\sqrt{2\pi}}\int_{-\infty}^\infty f(x)e^{-i\omega x}\ dx$
C. $\frac{1}{\pi}\int_{-\infty}^\infty f(x)e^{-i\omega x}\ dx$
D. None

126) Fourier transform generally applies to:
A. Periodic functions
B. Non-periodic functions
C. Even functions only
D. Odd functions only

127) F{f′(x)} equals (Where F represents Fourier Transform)
A. ωF(f(x))
B. −iωF(f)
C. ω^2F(f(x))
D. iωF(f(x))

128) Fourier Transform F{f(x − a)} equals: (Where F represents Fourier Transform)
A. $e^{-i\omega a}F(f)$
B. $e^{-\omega a}F(f)$
C. F(ω − a)
D. None

129) Convolution of f and g is defined as:
A. $\int_0^\infty f(t)g(x\ -\ t)\ dt$
B. fg
C. f + g
D. $\int_{-\infty}^\infty f(t)g(x\ -\ t)\ dt$

130) Fourier sine transform equals:
A. $\sqrt{\frac{2}{\pi}}\int_{-\infty}^\infty f(x)\ sin(\omega x)\ dx$

B. $\sqrt{\frac{2}{\pi}}\int_0^{\infty} f(x)e^{x}\, dx$

C. Σ f(x)

D. $\sqrt{\frac{2}{\pi}}\int_0^{\infty} f(x)\, sin(\omega x)\, dx$

131) Parseval's identity is:

A. $\int_{-\infty}^{\infty} |F(\omega)|^2\, d\omega \;=\; (\frac{1}{\sqrt{2\pi}})\ \int_{-\infty}^{\infty} |f(x)|^2\, dx$

B. ∫ f(x) dx = F(0)

C. F{f′} = iωF

D. None

132) If f(x) is odd, cosine coefficients are:

A. Zero

B. One

C. Infinite

D. None

133) Half-range cosine series is defined on:

A. (-L, L)

B. (0, L)

C. (1, ∞)

D. None

134) Fourier coefficient a_0 for f(x) = 1 on (-π, π) is:

A. 2π

B. 1

C. 2

D. ω

135) Fourier coefficient a_n for f(x) = 1 on (-π, π) is:

A. 2π

B. 1

C. 2

D. 0

136) Sine transform is useful for:

A. Odd extension

B. Even

C. Periodic

D. None

137) Cosine transform is useful for:

A. Even extension
B. Odd
C. Periodic
D. None

138) Fourier coefficient b_n for f(x) = c on (-π, π) is:
A. 2π
B. 1
C. 0
D. ω

139) Fourier transform converts:
A. Time domain to frequency domain
B. Frequency to time
C. Space to time
D. None

140) For f(x)=x on (−π, π), a_0 equals:
A. 1
B. π
C. 2π
D. 0

141) For f(x)=x on (−π, π), which coefficients are zero?
A. an
B. a_0
C. Both an and a_0
D. bn

142)Fourier sine coefficient of f(x)=x on (0, π) is:
A.0
B. 2/n
C. 1/n
D. $\frac{2(-1)^{\{n+1\}}}{n}$

143) Fourier Coefficient bn for f(x)=sin x on (−π, π):
A. 1 for n=1 only
B. 1 for all n
C. 0
D. −1

144)For f(x)=cos 3x on (−π, π), a_3 equals:
A. 3
B. 0

C. 1
D. π

145)Fourier series for $f(x)=x^3$ on (-2, 2) contains:
A. Cosine terms
B. Sine terms
C. Both
D. None

146)Fourier Transform of f(x−a) equals:
A. $e^{-i\omega a}$
B. F(ω−a)
C. $e^{-i\omega a}F\{f(x)\}$
D. None

147) What is the order and degree of the differential equation: $\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2 + y = 0$?
$A)\ Order\ 2, degree\ 2$
$B)\ Order\ 2, degree\ 1$
$C)\ Order\ 1, degree\ 4$
$D)\ Order\ 2, degree\ 4$

148) Which of the following is a homogeneous differential equation?
$A)\frac{dy}{dx} = \frac{x^2 + y^2}{xy}$
$B)\frac{dy}{dx} = x^3 + cos(y)$
$C)\ dy/dx = x + y^2$
$D)\frac{dy}{dx} = \frac{e^x}{y}$

149) The general solution of the equation $\frac{dy}{dx} = \frac{y}{x}$ is
$A)\ xy = C$
$B)\ y = C/x$
$C)\ y = C\,x^2$
$D)\ y = Cx$

*150)*For Bernoulli equation $\frac{dy}{dx} = 3x\,y + 4\,y^4$, the value of n is
$A)\ 4$
$B)\ 3$
$C)\ 1/3$
$D) -3$

151) If $F(f(x)) = \frac{1}{1+\omega^2}$, then $F\{f'\}$ equals:

A. $\frac{i\omega}{1+\omega^2}$

B. $-\frac{i\omega}{1+\omega^2}$

C. $\frac{\omega^2}{1+\omega^2}$

D. 0

152) Identify the correct integrating factor for $(1 + x^2)\frac{dy}{dx} - 2xy = 2x$

$A)\frac{1}{1 + x^2}$

$B)\ 1 + x^2$

$C)\ e^{\{-x^2\}}$

$D)\ e^{\{x^2\}}$

153) The general solution of $\frac{dy}{dx} + y = e^x$ is

$A)\ y = e^x + C$

$B)\ y = \frac{(e^x)}{2} + C\,e^{\{-x\}}$

$C)\ y = e^x\,(x + C)$

$D)\ y = Ce^x - e^x$

154) For the equation $\frac{dy}{dx} = y + xy^2$, let $v = \frac{1}{y}$. Then the transformed linear equation is

$A)\frac{dv}{dx} = -1 - xv$

$B)\frac{dv}{dx} + v = -x$

$C)\frac{dv}{dx} - v = x$

$D)\frac{dv}{dx} = v - x$

155) A Bernoulli equation is of the form

$A)\frac{dy}{dx} + P(x)y = Q(x)$

$B)\frac{dy}{dx} + P(x)y = Q(x)y^n$

$C)\frac{dy}{dx} = f(x/y)$

$D)\ M(x,y)\,dx + N(x,y)\,dy = 0$

156) The integrating factor for the linear equation $\frac{dy}{dx} + \frac{2}{x}y = x^2$ is

$A)\ e^x$

$B)\ e^{\{2\ ln\ x\}}\ =\ x^2$

$C)\ \frac{1}{x^2}$

$D)\ e^{\{x^2\}}$

157) Fourier coefficient a_0 for $f(x) = e^{-x};\ 0 \le x \le 2\pi$

A. $\frac{1}{\pi}\left(1-e^{-2\pi}\right)$

B. $\frac{1}{2\pi}\left(1-e^{2\pi}\right)$

C. $\frac{2}{\pi}\left(e^{-2\pi}-1\right)$

D. $\frac{1}{\pi}\left(1+e^{2\pi}\right)$

158) Fourier coefficient a_0 in the Fourier series $\frac{a_0}{2}+\sum_{n=1}^{\infty}\left(a_n \cos nx + b_n \sin nx\right)$ of $f(x) = x\sin x;\ 0 \le x \le 2\pi$ and $f(x+2\pi) = f(x)$ is

A 2

B 0

C -2

D -4

159) $f(x) = \begin{cases} 0 & -\pi < x < 0 \\ \cos x & 0 < x < \pi \end{cases}$ and f(x + 2π) = f(x). Fourier series of f(x) is represented by $\frac{a_0}{2}+\sum(a_n \cos nx + b_n \sin nx)$, then a_1 is

A $\frac{1}{2}$

B -1

C $-\frac{1}{2}$

D 1

160) The solution of the given differential equation: $y = (x-a)p - p^2,\ p = dy/dx$ is

(a) $y = (x-a)c - c$ (b) $y = (x-a)c + x^2$

(c) $y = (x-a)c - c^2$ (d) $y = (x-a)c^2$

161)The given differential equation $(y+x)dx-(y-x)dy=0$ is.

(a) Exact differential equation (b) Homogenous differential equation

(c) Both (a) and (b) (d) None of these

162) The integrating factor of the differential equation $(2xy)dx+(y^2-3x^2)dy=0$ is

(a) $-y^4$

(b) $\frac{1}{y^4}$

(c) $\frac{1}{y^2}$

(d) $-\frac{1}{y^4}$

163) The solution of the given differential equation: $p^2-7p+12=0$, which is solvable for p = dy/dx is

(a) $(y-4x-c)(y+3x-c)=0$ (b) $(y+4x-c)(y+3x-c)=0$

(c) $(y+4x-c)(y-3x-c)=0$ (d) $(y-4x-c)(y-3x-c)=0$

164) Which of the following functions is an integrating factor for the differential equation $(x^2+y^2+x)\,dx+xy\,dy=0$

(a) x (b) y

(c) y^2 (d) x^2

165) Which of the following equations is Clairaut's Equation?

(a) $y=py+\cos p$ (b) $y=py+\log p$

(c) $y=px+2x\log p$ (d) $y=px+\log p$

166) The orthogonal trajectory of the family of curves $\frac{dy}{dx}=x$ is given by

(a) $\frac{dy}{dx}=-x$ (b) $\frac{dy}{dx}=1$

(c) $\frac{dy}{dx}=-\frac{1}{x}$ (d) $\frac{dy}{dx}=-1$

167) In equation of the form $\frac{dy}{dx}+Py=Qy^n$, where P and Q are functions of x only or constants is known as……?

(a) Bernoulli's Equation (b) Cauchy-Euler differential equation,

(c) Clairaut's equation (d) None of these

168) The integrating factor of the linear differential equation $\frac{dy}{dx} + Py = Q$, where P and Q are functions of x only or may be constants.

(a) $e^{\int Qdx}$ (b) $e^{\int Pdx}$

(c) $e^{\int PQdx}$ (d) $e^{\int 2Pdx}$

169) The integrating factor of the linear differential equation $\frac{dy}{dx} + \frac{y}{x} = x^3$ is.

(a) x (b) y

(c) xy (d) None of these

170) Which of the following is exact differential equation.

(a) $(2xy + y^2)dx + (x^2 + 2xy)dy = 0$

(b) $(x^2y - 2xy^2)dx - (x^3 - 3yx^2)dy = 0$

(c) $(y^2 + xy)dx - x^2dy = 0$

(d) None of these

171) Integrating factor of the given Bernoulli's differential equation: $x\frac{dy}{dx} + y = x^3y^6$ is

(a) x^{-5} (b) x^5

(c) $5x^{-5}$ (d) $5x^5$

172) The solution of the differential equation $\frac{dy}{dx} + \frac{y}{x} = x^2$ is

(a) $xy = \frac{x^4}{4} + c$ (b) $\frac{y}{x} = \frac{x^3}{4} + c$

(c) $y = \frac{x^4}{4} + c$ (d) None of these

173) The solution of the differential equation $y' - y = e^{2x}$ is

(a) $c_1e^x + e^{2x}$ (b) $c_1xe^x + e^{2x}$

(c) $c_1e^x + xe^{-x}$ (d) $c_1e^x + e^{-2x}$

174) The solution of the differential equation $(x^2 - y)dx + (y^2 - x)dy = 0$ is

(a) $\frac{x^3}{3} - xy = c$ (b) $\frac{y}{x} = \frac{x^3}{4} + c$

(c) $\frac{x^3}{3} - xy + \frac{y^3}{3} = c$ (d) None of these

175) The general solution for the differential equation $y'' - 5y' + 6y = 0$ is

(a) $c_1e^{6x} + c_2e^{-x}$ (b) $c_1e^{3x} + c_2e^{2x}$

(c) $c_1e^{2x} + c_2xe^{-3x}$ (d) $c_1e^{3x} + c_2e^{-2x}$

176) The general solution for the differential equation $y'' - 3y' - 4y = 0$ is

(a) $c_1e^{4x} + c_2e^{-x}$ (b) $c_1e^{3x} + c_2e^{x}$

(c) $c_1e^{-4x} + c_2xe^{-x}$ (d) $c_1e^{-3x} + c_2e^{-x}$

177) What is the general solution of the Clairaut's equation $y = px + \frac{a}{p}$? where a is arbitrary constant.

a) $y = cx + \frac{a}{c}$
b) $y = c^2x + a$
c) $y = cx + ac$
d) $y = x/c + a/c$

178) What is the orthogonal trajectory of the family of circles $x^2 + y^2 = a^2$? a is arbitrary constant.

a) Parabolas $y^2 = 4ax$
b) Straight lines $y = mx$
c) Hyperbolas $xy = c$
d) Ellipses $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

179) In the auxiliary equation of a second-order linear ODE, if the roots are real and equal ($r_1 = r_2 = r$) the Complimentary solution is:

a) $C_1\, e^{rx} + C_2\, e^{rx}$
b) $(C_1 + C_2\, x)e^{rx}$
c) $e^{rx}(C_1 \cos x + C_2 \sin x)$
d) $C_1\, e^{rx} + C_2\, e^{-rx}$

180) In Variation of Parameters for $y'' + Py' + Qy = X$, the formula for the function $u_1(x)$ is:

a) $-\int \frac{y_2 X}{W}\, dx$ b) $\int \frac{y_2 X}{W}\, dx$
c) $\int \frac{y_1 X}{W}\, dx$ $d)\int -y_1 X dx$

181) If the equation $Mdx + Ndy = 0$ is homogeneous, the integrating factor is

a) $\frac{1}{Mx + Ny}$ b) $\frac{1}{Mx - Ny}$
c) $x^2 + y^2$ d) e^{x+y}

182) The total differential of $\frac{ydx - xdy}{y^2}$ is equal to

a) $d(xy)$ b) $d(x - y)$
c) $d(x + y)$ d) $d\left(\frac{x}{y}\right)$

183) The solution of the differential equation $ydx + xdy = 0$is

a) $xy = c$ b) $x = y + c$
c) $x - y = c$ d) None of these

184) The particular solution of the differential equation $y'' - 4y' + 4y = e^{-7t}$ is

(a) $y = \frac{e^{-7t}}{9}$ (b) $y = \frac{e^{7t}}{3}$

(c) $y = \frac{e^{2t}}{3}$ (d) $y = \frac{e^{-7t}}{81}$

185) The solution $y_1(x)$ & $y_2(x)$ are said to be L.I. if the wronskian (W)

(a) $W(y_1, y_2) = 1$ (b) $W(y_1, y_2) = 0$

(c) $W(y_1, y_2) \neq 0$ (d) $W(y_1, y_2) \neq 1$

186) The P.I. of $(D^2 + 4)y = \sin 2x$ is …

(a) $-\frac{x}{2}\cos 2x$ (b) $-\frac{x}{4}\cos 2x$

(c) $\frac{x}{2}\cos 2x$ (d) $\frac{x}{4}\cos 2x$

187)If the roots of the auxiliary equation are 3, 3, 3 the Complementary Function (C.F.) is:

a) $(C_1 + C_2x)e^{3x}$ (b) $C_1 e^{3x} + C_2e^{3x} + C_2e^{3x}$

(c) $(C_1 + C_2x + C_3x^2)e^{3x}$ (d) $C_1 e^{3x} + C_2e^{-3x}$

188) The general solution of the differential equation $y'' - 2y' + 2y = 0$ is

(a) $e^x(C_1 \cos 2x + C_2 \sin 2x)$ (b) $e^{2x}(C_1 \cos 2x + C_2 \sin 2x)$

(c) $e^x(C_1 \cos x + C_2 \sin x)$ (d) $(C_1 + C_2x) e^x$

189) The solution of the differential equation $(D^2 - 6D + 8)y = e^{6x}$, $D = \frac{d}{dx}$

(a) $(C_1e^{2x} + C_2e^{4x}) - \frac{e^{6x}}{8}$

(b) $(C_1e^{2x} + C_2e^{4x}) + \frac{e^{6x}}{8}$

(c) $(C_1e^{2x} + C_2e^{4x}) + \frac{e^{6x}}{80}$

(d) $(C_1e^{6x} + C_2e^{4x}) + \frac{e^{6x}}{8}$

190) In the method of Variation of Parameters, the Wronskian W for complementary solution of $y'' + y = \sec x$ is:

(a) 0

(b) 1

(c) $\sin x$

(d) $\cos x$

191) The particular integral of the differential equation $(D^2 + 9)y = \sin 2x$ is

(a) $-\frac{1}{5}\sin 3x$ (b) $-\frac{1}{5}\sin 2x$

(c) $\frac{1}{5}\sin 2x$ (d) $\frac{1}{5}\sin 3x$

192) Which is the correct assumption of P.I. for $y'' - 4y' = 6x^2$.

(a) $Ax + B$ (b) Ax^2

(c) $Ax^2 + Bx$+c (d) $Ax^3 + Bx^2 + Cx$

193) **T**he Wronskian (W) of the functions $1, e^x$?

(a) e^x (b) $1 + e^x$

(c) 0 (d) e^{-x}

194) Which method can be best suitable to find the complete solution of the differential equation $y'' + y = \tan x$?

(a) By formula for particular integral (b) Variation of Parameters

(c) Method of undetermined coefficients (d) Homogeneous method

195) The solution of the differential equation $(D^2 + 9)y = 0$ is

(a) $(C_1 \cos 3x + C_2 \sin 3x)$ (b) $e^{3x}(C_1 \cos 3x + C_2 \sin 3x)$

(c) $e^x(C_1 \cos 3x + C_2 \sin 3x)$ (d) $(C_1 + C_2 x)\, e^{3x}$

196) If $f(D)y = e^{ax}$ and $f(a) = 0$, then particular integral y_p involves a factor of:

(a) x

(b) x^2

(c) $ln\, x$

(d) e^x

197) The assumption of P.I. for $(D-2)^2 y = 17e^{2x}$.

(a) Ae^{2x} (b) Axe^{2x}

(c) Ax^2e^{2x} (d)) $Axe^{2x} + x^2$

198) The roots of the auxiliary equation of $D^2y + Dy + y = 0$ are:

(a) Real and equal

(b) Real and distinct

(c) Complex conjugates

(d) Zero

199) The solution of the differential equation $(x + y - 10)dx + (x - y - 2)dy = 0$ is

(a) $x^2 + 2xy - 20x - y^2 - 4y = c$

(b) $x^2 + xy - 10x - y^2 - 2y = c$

(c) $x^2 + xy - 10x - y^2 - 2y = c$

(d) $x^2 + 2xy - 10x - y^2 - 4y = c$

200) The solution of the Clairaut's equation i.e., $p = \log(px - y)$, where p is dy/dx.

(a) $y = \log(cx - y)$ (b) $y = cx - e^c$

(c) $y = log(-cx - y)$ *(d)* $y = cx - e^p$

201) What are the orthogonal trajectories of the family of curves: $y = ax^2$, where a, b are parameter?

(a) $\frac{x^2}{2} + y^2 = b$ (b) $\frac{x^2}{2} - y^2 = b$

(c) $\frac{x^2}{2} - y^3 = b$ (d) $\frac{x^3}{2} + y^2 = b$

202) Solve $(x^2 - ay)dx = (ax - y^2)dy$. The solution is ....

(a) $x^3 - 3axy + y^3 = C$ (b) $x^3 - 3axy - y^3 = C$

(c) $x^3 - y^3 = C$ (d) $x^3 + y^3 = C$

203) The integrating factor of the differential equation $(x + 2)\frac{dy}{dx} - y = e^x(x + 2)^2$ is

(a) $x + 2$ (b) $\frac{1}{x+2}$

(c) $(x + 2)^2$ (d) $\frac{1}{(x+2)^2}$

204) The solution of the differential equation $(2xy + y^2)dx + (x^2 + 2xy)dy = 0$ is

(a) $x^2y + y^2 = c$ (b) $x^2y - xy^3 = c$

(c) $x^2y + xy^2 = c$ (d) None of these

205) The integrating factor of the differential equation $y(y^2 - 2x^2)dx + x(2y^2 - x^2) = 0$ is

(a) $\frac{1}{y^2-x^2}$ (b) $\frac{1}{3xy(y^2-x^2)}$

(c) $\frac{1}{2xy(y^2-x^2)}$ (d) $\frac{1}{2(y^2-x^2)}$

206) The orthogonal trajectories for the one-parameter family of curve $x^2 + 4y^2 = c$ will be

a) $y = x^4 + c$ (b) $y = x^2 + c$

(c) $y = x^5 + c$ (d) $y = x + c$

207) The general solution of the defined differential equation: $(D^2 - 3D + 2)$y=e^{3x}, where $D = \frac{d}{dx}$?

(a) $y = c_1e^x - c_2e^{2x} - \frac{e^{3x}}{2}$ (b) $y = c_1e^x + c_2e^{2x} + \frac{e^{3x}}{2}$

(c) $y = c_1e^{2x} + c_2e^{2x} + \frac{e^{3x}}{2}$ (d) $y = c_1e^{2x} - c_2e^{2x} - e^{3x}$

208) The particular integral of the differential equation $(D^2 + 4) = sin3x$ is

(a) $-\frac{1}{5}\sin 3x$ (b) $-\frac{1}{6}\sin 3x$

(c) $\frac{1}{6}\sin 3x$ (d) $\frac{1}{5}\sin 3x$

209) On solving the differential equation $y'' + y = \sec x$ using the method of variation of parameters if the particular solution is assumed as $uy_1 + vy_2$, then the value of the function u is

(a) $\tan x$ (b) $\ln|\sin x|$

(c) $\ln|\cos x|$ (d) x

210) The complete solution of the differential equation $(D^2 + 4)y = 16$ is

(a) $(C_1 \, cos\, 2x + C_2 sin\, 2x) + 4$ (b) $(C_1 \, cos\, 2x + C_2 sin\, 2x)$

(c) $(C_1 \, cos\, 2x + C_2 sin\, 2x) + 16$ (d) $e^{2x}(C_1 \, cos\, x + C_2 sin\, x) + 4$

211) The particular integral of the differential equation $\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 9y = e^{3x}$

(a) e^{3x} (b) $\frac{x^2e^{3x}}{2}$

(c) xe^{3x} (d) $\frac{xe^{3x}}{2}$

212) The solution of the differential equation $\frac{d^2y}{dx^2} + 6\frac{dy}{dx} + 9y = 5e^{3x}$ is

a) $(c_1 + xc_2)e^{-3x} + \frac{e^{3x}}{36}$ b) $(c_1 + xc_2)e^{3x} + \frac{5e^{3x}}{16}$

c) $(c_1 + xc_2)e^{-3x} + \frac{5e^{3x}}{36}$ d) $(c_1 + xc_2)e^{3x} + \frac{e^{3x}}{16}$

213) The particular integral of the differential equation $\frac{d^2y}{dx^2} + 4y = e^{-2x}$

(a) $\frac{e^{-2x}}{16}$ (b) $\frac{e^{-2x}}{8}$

(c) $\frac{e^{-2x}}{2}$ (d) $\frac{-e^{-2x}}{2}$

214) The particular integral of $(D^2 - 1)y = x$ is:

(a) x

(b) -x

(c) x^2

(d) 1

215) The solution of the differential equation $xdy - ydx + 2x^3dx = 0$ is

(a) $\frac{y}{x} + x^2 = c$ (b) $\frac{y}{x} - \frac{x^2}{2} = c$

(c) $xy = c$ (d) None of these

216) The solution of the differential equation $\frac{d^2y}{dx^2} + 8\frac{dy}{dx} + 16y = 0$ is

a) $(c_1 + xc_2)e^{4x}$ b) $c_1e^{-4x} + c_2e^{-4x}$

c) $(c_1 + xc_2)e^{-4x}$ d) $(c_1 + xc_2)e^{8x}$

217) The solution of the differential equation $(D^2 + 9)y = \sin 4x$ is:

a) $y = c_1 \cos 3x + c_2 \sin 3x + \sin 4x$

b) $y = c_1 \cos 3x + c_2 \sin 3x - \frac{1}{7}(\sin 4x)$

c) $y = c_1 \cos 3x + c_2 \sin 3x + \frac{1}{7}(\sin 4x)$

d) $y = c_1 \cos 3x + c_2 \sin 3x - 7(\sin 4x)$

218)The orthogonal trajectory of $e^x + e^{-y} = c$ is

a) $e^x + e^y = c$

b) $e^y - e^{-x} = c$

c) $e^{-x} - e^{-y} = c$

d) $2e^x + e^{-5y} = c$